

Lecture 3 $\hookrightarrow$

* Think about some possible binary¹ classification, multiclass² classification and regression³ tasks. What hypothesis space can be considered?

□ hypothesis space (H) is the set of all the possible legal hypothesis. This is the set from which the machine learning algorithm would determine the best possible (only one) which would best describe the target function or the outputs. $\Rightarrow$ (best solution = hypothesis) for finding the best possible we need to find the right balance between expressivity and simplicity. So, we should start by exploring the dataset, using visualization as much as possible.

The key is that, the algorithm with too expressive spaces can generalize poorly to unseen data and be too complex to understand, whereas those with overly simple hypothesis may underfit the data.

$\perp$ binary classification $\hookrightarrow$ it depends on how data spread in our space.

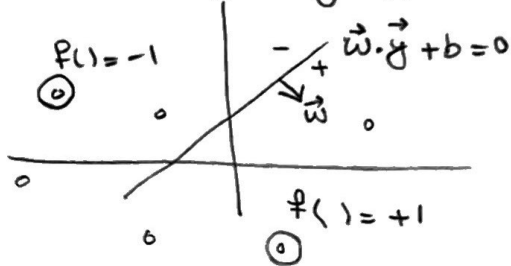
For instance we can conclude that a straight line isn't likely to be an adequate boundary for the classification data. However a high-order curve would probably be too complex even though it might split the dataset into two classes without an error.

for example our hypothesis space can be $\Rightarrow$

[1] hyperplane in $\mathbb{R}^2$:

$$X = \{y \mid y \in \mathbb{R}^2\}$$

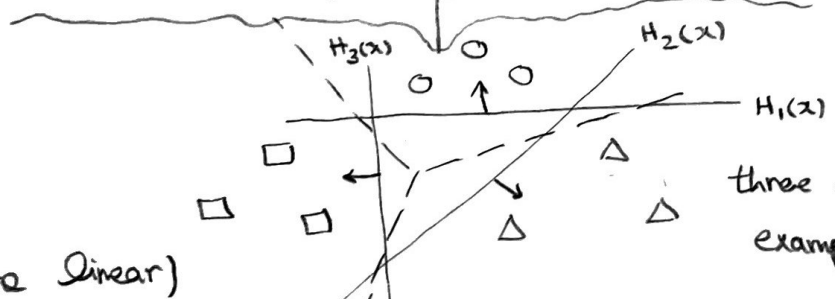
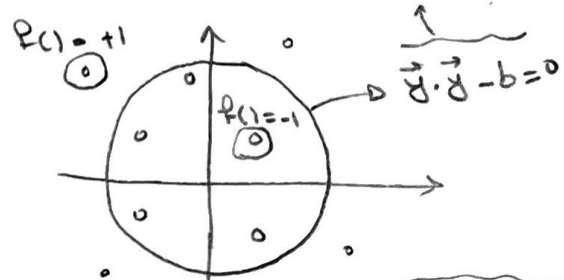
$$H = \{f_{w,b}(y) = \text{sign}(w \cdot y + b), w \in \mathbb{R}^2, b \in \mathbb{R}\}$$



[2] circle in $\mathbb{R}^2$:

$$X = \{y \mid y \in \mathbb{R}^2\}$$

$$H = \{f_b(y) = \text{sign}(\|y\|^2 - b), b \in \mathbb{R}\}$$



$\perp$ multiclass classification $\hookrightarrow$

one example $\hookrightarrow$

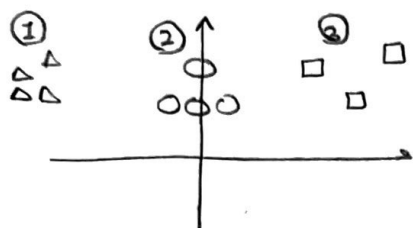
(The hypothesis space can be linear)

we have two methods to consider $\rightarrow$

- 1. one-vs-all $\rightarrow$ Train K binary classifier, one for each class
- 2. one-vs-rest $\rightarrow$ train i th classifier to distinguish class i from rest

$\rightarrow$ base hypothesis space is $H = \{h: X \times Y \rightarrow \mathbb{R}\}$

For example for three class $\rightarrow$



base hypothesis space: $H = \{f(x) = w^T x \mid x \in \mathbb{R}^2\}$

$\rightarrow$ separating boundary always contains the origin

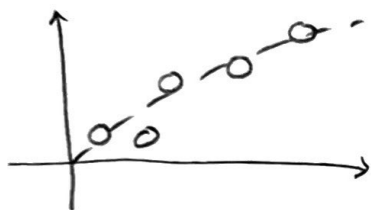
3. regression $\rightarrow$

linear regression can be limited to linear functions as its hypothesis space, or it can be expanded to learn polynomials.

$\rightarrow$ for example for polynomial in $\mathbb{R}^2$ $\rightarrow$

$$H = \{f_{w,b}(y) = \vec{w} \cdot y + b, w \in \mathbb{R}^2, b \in \mathbb{R}\}$$

$\rightarrow$ = second degree polynomial



- but in general $\Rightarrow$ hypothesis space: that consists the set of linear functions (hyperplanes) $\rightarrow h(x) = w_0 + w_1 x_1 + \dots + w_k x_k$

